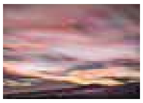
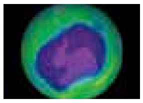


Nacreous clouds over Sweden
These nacreous clouds were captured earlier this year, and the colours look out of this world. **https://dgit.in/may20-57**



Ozone layer science
Learn about chemical reactions that take place in the ozone layer, including ones with chlorine. **https://dgit.in/may20-58**

Even the lower stratospheric clouds show iridescence, but the iridescence is far more vibrant and vivid in the nacreous clouds, and even most rainbows are duller in comparison. The colours of the thin wispy clouds, that roll, coil and uncoil are so bright, that they are sometimes mistaken for aurorae, but the two are totally unrelated atmospheric phenomena. There are some indications that the clouds are formed after violent tropical storms throw up water vapour into the stratosphere.

Depending on the chemical composition of the clouds, the nacreous clouds are classified into two types. These are type I and type II. Type II clouds are mostly made up of water ice. Type I is further classified into three subtypes, type Ia, type Ib and type Ic. Type Ia is made up of large droplets consisting of a compound known as nitric acid trihydrate, type Ib is made up of small droplets containing a mixture of sulfuric acid, nitric acid as well as water, and type Ic is made up of nitric acid and water. The exact chemical composition of these clouds can be accurately ascertained through LIDAR observations. Despite decades of research on these rare clouds, the process of their formation, and their periodicity is little understood. Some institutions and researchers such as the Australian Antarctic Division consider only the type II PSCs to be nacreous clouds.

THE DARK SIDE

For the longest time, PSCs were considered to be curiosities that didn't really have any impact on the planet. However, as scientists learned more about them in the last few decades, the clouds have an impact on the environment, and not in a good way. While type II clouds are benign, the type I clouds have a role to play in the depletion of the ozone layer.

When temperatures fall below (-110°C) chlorine nitrate and hydrochloric acid in the stratosphere can interact with the type I PSCs and start a chemical reaction that is destructive to the ozone layer, converting the ozone to oxygen. While ozone is harmful to humans if breathed in, in the atmosphere it plays an important role of shielding the life forms on Earth from the ravages of the Sun, particularly in the form of ultraviolet light. It is this layer that bounces off the heat and energy from the Sun, causing the temperatures in the stratosphere to increase with altitude. For years, humans were pumping chlorofluorocarbons (CFCs) into the atmosphere through the indiscriminate use of refrigerators, air conditioners and aerosol sprays. These chemicals accumulated in the ozone layer, leading to an accumulation of chlorine and bromine, which also interact with the nacreous clouds to further destroy the ozone. The clouds provide a platform for these destructive chemical reactions to take place. If it were not for these clouds, the destruction of the ozone layer would be considerably reduced. Since these nacreous clouds are formed more frequently over the Antarctic region, they were responsible for the formation of the ozone hole over the south pole. The nacreous clouds also play a role in the seasonal appearance of the ozone hole over the South Pole, because every time the temperature drops below the frost point, the chemicals begin their destructive reactions using the newly formed nacreous clouds as a nucleus for their activity.

While the Montreal Protocol signed in 1987, along with its amendments have greatly reduced the use of CFCs starting from 1996, the particles already in the atmosphere are expected to remain there for 50 to 100 years. These CFCs continue to interact with the nacreous clouds, eroding the ozone layer. Just a single atom of chlorine can convert 100,000 ozone molecules into the oxygen that we breathe, before disappearing from the atmosphere. That rate is far higher than the rate at which ozone can be naturally replenished, and should give an idea of exactly how damaging the chemicals that the industrial society has thoughtlessly deposited into the planet are. Apart from the CFCs, there are also other ozone-depleting substances (ODS) such as halons and methyl bromide, which contain bromine. Bromine, like chlorine, is destructive to the ozone layer, and also uses the nacreous clouds as a platform.

One of the aggravating factors is the complex interactions of energy and chemicals in the atmosphere, the area between the ozone layer and the lower stratosphere is critical in maintaining a delicate balance. The emissions of greenhouse gases by industrial activity are heating the atmosphere, by trapping more heat in the troposphere and preventing them from escaping. As a result, the lower regions of the stratosphere are cooler than they otherwise would have been, leading to temperatures in this layer dipping below the frost point, and forming more nacreous clouds. Because of this reason, nacreous clouds are being spotted more frequently, and in lower latitudes than previously seen.

Credit: P-M Hedén/APOD

Nacreous clouds over Sweden